



How Flight Prices Vary Across Airline, Classes, and Booking Windows in India – Part Two: Regression Prediction Modeling

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Introduction

- Airline tickets prices in India are widely vary based on the factors (time of bookings, airlines, class, number of stops)
- Knowing this patterns helps travellers make decisions and airline optimize pricing
- Contains over 300,000 records with flight information with airline, class, stops, days until departure and price
- Goal of this project to explore how different factors influence prices in India, finding trends and provide actionable insights
- Conduct SAS Software for statistical analysis and regression modelling

Where did I get the Datasets?

kaggle

- Got datasets from Kaggle's source
- The datasets named is "*Flight Price Prediction*"

Plan for this Part (Part Two)

Part One used for Tableau to create charts and dashboards

Part Two conduct with SAS Software to do Statistical Modelling and Regression Modelling



Variables Description

Dependent Variable	Descriptions
price	Price for the flight
Independent Variables	
airline (airline_num)	List of Airline (Air India, Vistara, Indigo, Air Asia, etc)
stops (num_stops)	Number of Stops
destination_city (dest)	City of Destination
class (classes / class_business / class_economy)	Types of Class (Economy / Business)
days_left	Number of Days before departure

Regression Equation

- **Given Equation:**

- $Y_i = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n + \varepsilon_i$

- **Our Equation:**

- $price = \beta_0 + \beta_1 airline_1 + \beta_2 stops_2 + \beta_3 destination_city_3 + \beta_4 class_4 + \beta_5 days_left_5 + \varepsilon_i$

Hypothesis Testing



- *Null Hypothesis (H_0): There is no significant in predicting the price based on the factors*
- *Alternate Hypothesis (H_1): There is significant in predicting the price based on the factors*

The First 10 Observations of Original Datasets

```
proc print data=updated_data (obs=10);  
title 'The first 10 updated dataset';
```

SAS Code for “The First 10 Observations of Original Datasets”

The first 10 original datasets

Obs	VAR1	airline	flight	source_city	departure_time	stops	arrival_time	destination_city	class	duration	days_left	price
1	0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai	Economy	2.17	1	5953
2	1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.33	1	5953
3	2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	Mumbai	Economy	2.17	1	5956
4	3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.25	1	5955
5	4	Vistara	UK-963	Delhi	Morning	zero	Morning	Mumbai	Economy	2.33	1	5955
6	5	Vistara	UK-945	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.33	1	5955
7	6	Vistara	UK-927	Delhi	Morning	zero	Morning	Mumbai	Economy	2.08	1	6060
8	7	Vistara	UK-951	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.17	1	6060
9	8	GO_FIRST	G8-334	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.17	1	5954
10	9	GO_FIRST	G8-336	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.25	1	5954

**Need to check the
original datasets for
the first 10
Observation before
started to do
Regression Modelling**

SAS Output for “The First 10 Observation of Original Datasets”

All Category Variable will convert into Numerical Variable to run Regression Modelling

```
data updated_data;
set flight_prices_india;
if airline = 'Vistara' then airline_num = 1;
if airline = 'Air India' then airline_num = 2;
if airline = 'SpiceJet' then airline_num = 3;
if airline = 'GO_FIRST' then airline_num = 4;
if airline = 'Indigo' then airline_num = 5;
if airline = 'AirAsia' then airline_num = 6;

if stops = 'zero' then num_stops = 0;
if stops = 'one' then num_stops = 1;
if stops = 'two' then num_stops = 2;

if destination_city = 'Bangalore' then dest = 1;
if destination_city = 'Chennai' then dest = 2;
if destination_city = 'Delhi' then dest = 3;
if destination_city = 'Hyderabad' then dest = 4;
if destination_city = 'Kolkata' then dest = 5;
if destination_city = 'Mumbai' then dest = 6;

if class = 'Busines' then classes = 1;
if class = 'Economy' then classes = 2;

run;
```

SAS Code for covert all Category Variables into Numerical Variables

To run Regression Modelling, it must be numerical variables. And based on this datasets, few of the variables are categorical and need to convert into numerical to run Regression. Otherwise, when trying to run Regression with categorical variables, it will end up with error and may not go through.

The First 10 Observation of Updated Datasets

```
proc print data=updated_data (obs=10);  
title 'The first 10 updated dataset';
```

Now, this datasets are all numerical variables, so it will be ready to do regression modelling.

SAS Code for the First 10 Observation of Updated Datasets

The first 10 updated dataset

Obs	VAR1	airline	flight	source_city	departure_time	stops	arrival_time	destination_city	class	duration	days_left	price	airline_num	num_stops	dest	classes
1	0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai	Economy	2.17	1	5953	3	0	6	2
2	1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.33	1	5953	3	0	6	2
3	2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	Mumbai	Economy	2.17	1	5956	6	0	6	2
4	3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.25	1	5955	1	0	6	2
5	4	Vistara	UK-963	Delhi	Morning	zero	Morning	Mumbai	Economy	2.33	1	5955	1	0	6	2
6	5	Vistara	UK-945	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.33	1	5955	1	0	6	2
7	6	Vistara	UK-927	Delhi	Morning	zero	Morning	Mumbai	Economy	2.08	1	6060	1	0	6	2
8	7	Vistara	UK-951	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.17	1	6060	1	0	6	2
9	8	GO_FIRST	G8-334	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.17	1	5954	4	0	6	2
10	9	GO_FIRST	G8-336	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.25	1	5954	4	0	6	2

SAS Output for the First 10 Observation of Updated Datasets

Regression Modelling

```
* Regression Modelling;  
proc reg data = updated_data;  
    model price = airline_num num_stops dest classes days_left;  
    title 'Regression Model for Average Price Based on the Factors';  
run;
```

SAS Code for Regression Modelling

Regression Model for Average Price Based on the Factors

The REG Procedure
Model: MODEL1
Dependent Variable: price

Number of Observations Read	300153
Number of Observations Used	82405
Number of Observations with Missing Values	217748

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	3.874667E13	7.749335E12	150006	<.0001
Error	82399	4.256752E12	51660241		
Corrected Total	82404	4.300343E13			

Root MSE	7187.50588	R-Square	0.9010
Dependent Mean	19022	Adj R-Sq	0.9010
Coeff Var	37.78500		

Results from the Output:

- R-Squared = 0.9010 (90.10%)
- Adjust R-Squared = 0.9010 (90.10%)
 - Shows an Excellent Model Fit
 - Predictors are meaningful
 - Suggest that the model is not overfitted

Testing:

Since the Pr > F result: <0.001 and it is less than alpha = 0.05, do not reject the null hypothesis and conclude that this model it is proven significant that price is depending on the factors.

SAS Output for Regression Modelling (Part One)

Regression Modelling (Continued)

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	96493	148.11895	651.46	<.0001
airline_num	1	-355.49930	15.39379	-23.09	<.0001
num_stops	1	8192.39549	65.79045	124.52	<.0001
dest	1	404.70180	16.74447	24.17	<.0001
classes	1	-47000	65.71229	-715.23	<.0001
days_left	1	-138.64953	1.85673	-74.67	<.0001

SAS Output of Regression Modelling (Part Two)

Interpret the Parameter Estimate:

- Intercept = 96493
 - Baseline value when all variables = 0
- airline_num = - 355.49930
 - Price decreases by estimated of 355 if airlines increases
 - Some airlines are cheaper than others
- num_stops = 8192.39549
 - Price increases by estimated of 8193
 - Usually, more stops = cheaper flights
 - Flights with more stops may be longer or premium routes
- dest = 404.70180
 - Price increases by estimated of 405
 - Some of destinations are more expensive (Mostly popular destinations)
- classes = -47000
 - Price reduced by 47000 if switching to different class
 - Looks very strong effect
 - But will conduct another regression testing with Business and Economy
- days_left = -138.64953
 - Price decreases by estimated of 139
 - If booking earlier, will get cheaper flights

Regression Modelling (Continued)

Results from the Predictor

Based on each predictors, since the p-value < 0.0001 and it is less than $\alpha = 0.05$, do not reject the null hypothesis and conclude that Price is depending on the factors.

Flight results: 20 flights found		Economy	Premium Economy	Business Class
19:25	1 stop 2hr 58m	20:21		
Toronto				
YUL				
= 1hr 25m				
Vancouver				
included travel operated by Air Canada Express - jax				
Seats Details				
		\$350	\$956 Mixed cabin	\$1369 Includes lie-flat seats
Flight results: 18 flights found		Economy Class	Premium Economy	Business Class
15:25	1 stop 7hr 58m	20:23		
Toronto				
YUL				
= 1hr 25m				
Vancouver				
Included travel operated by Air Canada Express - jax				
Premium seats Details				
		2 seats left 16.4K + CA 101	1 seat left 46.7K + CA 101 Mixed cabin	1 seat left 60K + CA 101 Mixed cabin



Why it is better to make Separate Regressions for Business and Economy Classes?

It would be able to show consistent results with both variables remaining statistically significant. However, change in coefficient signs reflect differences in reference categories rather than change in the underlying relationship, confirming that ticket class has a strong and stable impact on price. However, when I run the regression model (as shown in the next two-three slide/page) all the results are the same except Business and Economy Class as Business is positive 47000 while economy is negative 47000.

Creating Dummy Variables for Flight Classes

```
* Create a dummy variables for flight classes;  
if class = 'Business' then class_business = 1;  
else class_business = 0;  
  
if class = 'Economy' then class_economy = 1;  
else class_economy = 0;  
run;
```

SAS Code for create dummy variables for Flight Classes

Before running a Regression for only with Business and Economy Class, I created dummy variables where class_business = 1 if class is Business, otherwise class_business = 0, and class_economy = 1 if class is Economy, otherwise class_economy = 0.

Regression Modelling (Only Business Class)

```
proc reg data = updated_data;  
    model price = airline_num num_stops dest class_business days_left;  
    title 'Regression Model for Average Price Based on the Factors for Business Class';  
run;
```

SAS Code for Regression (Only Business Class)

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	2494.28121	120.79099	20.65	<.0001
airline_num	1	-355.49930	15.39379	-23.09	<.0001
num_stops	1	8192.39549	65.79045	124.52	<.0001
dest	1	404.70180	16.74447	24.17	<.0001
class_business	1	47000	65.71229	715.23	<.0001
days_left	1	-138.64953	1.85673	-74.67	<.0001

Business Class (class_business) = 47000

- Price increased by 47000
- Explains that Business Class is more expensive
- Since P-value < 0.001 and less than alpha = 0.05, do not reject the null hypothesis and conclude that it is significant and explains Business Class is more expensive.

SAS Output for Regression (Only Business Class)

Regression Modelling (Only Economy Class)

```
proc reg data = updated_data;  
    model price = airline_num num_stops dest class_economy days_left;  
    title 'Regression Model for Average Price Based on the Factors for Economy Class';  
run;
```

SAS Code for Regression Modelling (Only Economy Class)

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	49494	118.09631	419.10	<.0001
airline_num	1	-355.49930	15.39379	-23.09	<.0001
num_stops	1	8192.39549	65.79045	124.52	<.0001
dest	1	404.70180	16.74447	24.17	<.0001
class_economy	1	-47000	65.71229	-715.23	<.0001
days_left	1	-138.64953	1.85673	-74.67	<.0001

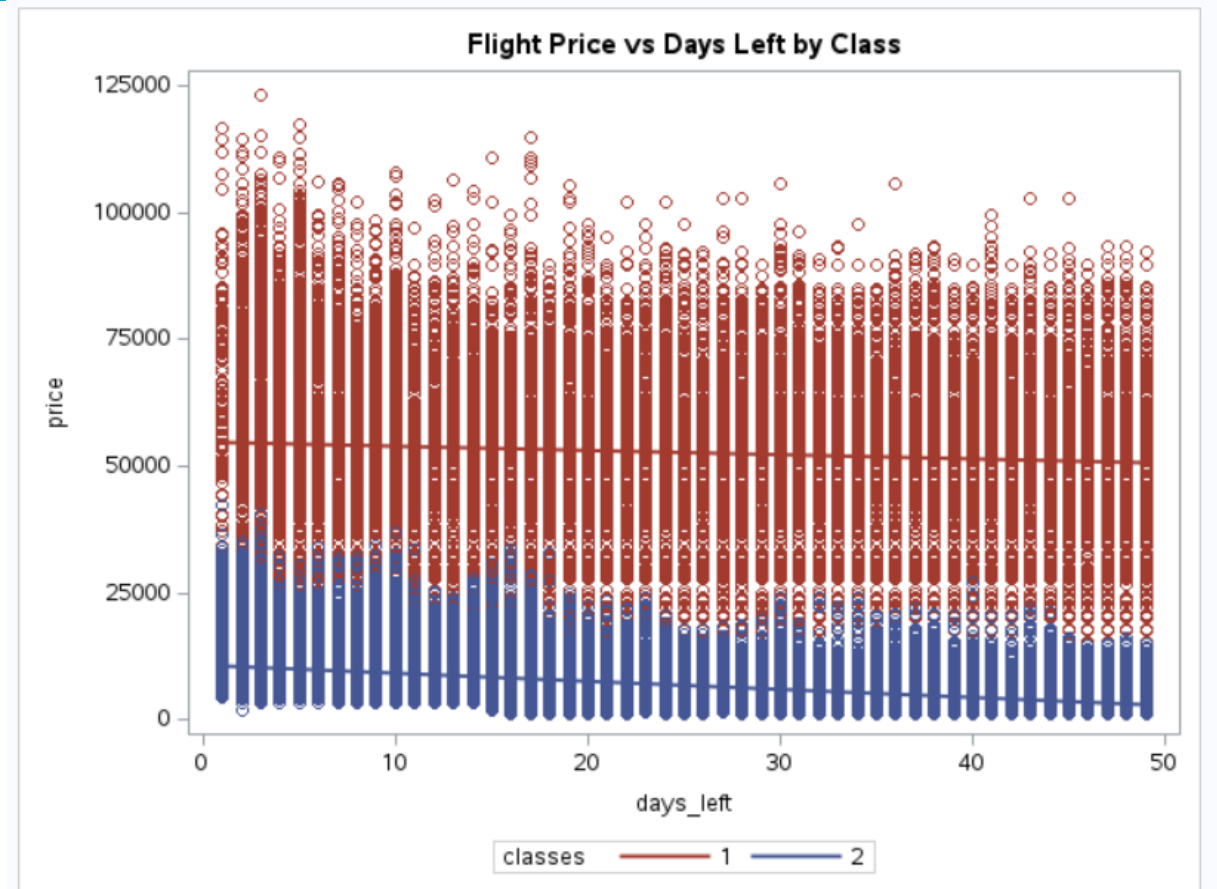
class_economy (Economy Class) = -47000

- Price decrease by 47000
- Since p-value < 0.0001 and less than alpha = 0.05, do not reject the null hypothesis and conclude that it is significant and shows booking Economy is cheaper
- Economy Class is cheaper

SAS Output for Regression Modelling (Only Economy Class)

Flight Price Distribution by Class and Days Left

- Class 1 (Red) = Business
- Class 2 (Blue) = Economy
- Business Class prices shows higher than Economy Prices across all days left
- Business Class shows more spread out vertically due to greater price variability
- Slight decreasing trend in price for both classes as days gets later
- Outliers exist in Business class, with some extremely high prices
- Airline class shows strong determinant of flight price, and last-minute bookings tends to be more expensive, especially booking with Business Class



Conclusion

- **Model Validation:** Regression shows excellent model with the flight prices based on $R\text{-Squared} = 0.9010$. It shows an excellent model fit and selected factors are meaningful predictors of price
- **Overall Significance:** Model shows statistically significant ($p\text{-value} < 0.0001$), proving that flight prices in India are dependent on the combination of factors.
- **Impact of Ticket Class:** Class is the strongest determinant of price
 - **Business Class:** Price Increase of \$47,000
 - **Economy Class:** Price Decrease of \$47,000
 - **Scatter Plot** confirms Business Class prices are consistently higher and more variable than Economy Prices across all booking windows
- **Impact of Booking Window:** `days_left` have negative coefficient (-138.65) shows booking earlier may have cheaper prices. However, booking in the last minutes can cause higher prices.
- **Other Factors:** airline, number of stops, and destination city are also statistically significant factors that influence the ticket price.



Recommended for Next Step

- Testing for Interaction Effects (class * days_left)
- Research Question for recommendation in the next step:
 - Does the number of days before departure affect Business and Economy class prices differently?
- It separates the baseline class premium from the last-minute booking penalty
- Reveals if Business class fares are more sensitive to booking window than Economy
- Able to provide more precise guidance for travellers and airlines



Source

- <https://www.kaggle.com/datasets/shubhambathwal/flight-price-prediction>